

Georgios Nicolaides

Cybersecurity Engineer

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OBJECTIVE

Highly motivated MEng Cybersecurity graduate with a strong technical foundation in protocol security and cryptographic optimization. I am eager to launch a career in a fast-paced environment where I can contribute my analytical skills while actively developing my professional expertise. Committed to delivering results, supporting team objectives, and leveraging my background to drive security initiatives.

EDUCATION

University of Limerick, Ireland - *Master of Engineering Cybersecurity*

SEP 2025 - AUG 2026

Harokopio University, Greece - *Bachelor of Science Informatics & Telematics*

OCT 2021 - JUL 2025

SKILLS

Technical Core & Engineering

- **Security Engineering:** Secure SDLC integration, Static Code Analysis, Cryptographic implementation (AES, RSA, Merkle Schemes, applied in LEO satellite authentication framework).
- **System Administration:** Hardening Linux/Unix environments, Infrastructure as Code (Ansible, Docker, Kubernetes).
- **Programming:** Proficient in C, C++, Java, Javascript and Python.
- **Cloud & Infrastructure Security:** Foundational knowledge of cloud security principles across AWS/Azure environments, including access control and secure configuration.

Cybersecurity Operations & Analysis

- **Defensive Security:** Log Analysis, SIEM deployment (ELK Stack, Splunk), Incident Monitoring, and Firewall/ACL management.
- **Offensive Security:** Vulnerability assessment and network/host enumeration (Nmap, Wireshark, Burp Suite).
- **Compliance & Risk:** Working knowledge of NIST Cybersecurity Framework and GDPR data protection requirements, applied in security assessments and tool development.

Hardware & Embedded Systems

- **IoT Security:** Specialisation in RFID/NFC (MIFARE) security and hardware interfacing for resource-constrained devices, including prototype development and adversarial testing.

Leadership & Communication

- Co-led a university developer community, scaling active membership from 5 to 50; mentored 30+ students across competitive cybersecurity environments, delivering monthly technical workshops on offensive and defensive topics.

VOLUNTEERING EXPERIENCE

Google Developer Group (GDG) Sub-chapter, Harokopio University *Co-Lead (2024–2025) | Core Team Member (2023–2024)*

- Revived a dormant developer community, growing active membership by transitioning to a GDG sub-chapter model.
- Delivered monthly cybersecurity workshops covering SQL Injection, network enumeration, privilege escalation, and OSINT.
- Mentored 6 teams (30+ participants) through the Hellenic University Hack competition; all secured top-20 finishes with no prior cybersecurity experience.
- Secured a partnership with the National & Kapodistrian University GDG to preserve the community through institutional restructuring.

KEY PROJECTS

MEng Thesis Research | University of Limerick

Project: Securing RFID/NFC Technologies in Resource-Constrained IoT Environments (2026)

- **Design & Development:** Engineered the **SONDA-MA protocol**, a lightweight mutual authentication mechanism for passive RFID/NFC hardware.
- **System Integration:** Developed a prototype using Raspberry Pi 5, MFRC522/PN532 modules, and implemented rigorous adversarial testing with Flipper Zero.
- **Impact:** Achieved verifiable replay protection and dynamic session management within Class 1 IoT hardware constraints, bridging the gap between theoretical security and physical deployment feasibility.

Research Paper (First Author) | Pending Submission

Project: LEO Satellite IoT Authentication Protocol (2026)

- **Cryptographic Optimization:** Architected a high-performance authentication framework for LEO satellite-to-ground communication.
- **Performance Engineering:** Optimized C-based protocol logic, achieving a **1,807x improvement** in key generation speed compared to traditional RSA-2048, significantly reducing latency and energy overhead for constrained satellite endpoints.
- **Simulation & Testing:** Developed a containerized Docker-based testbed to validate mutual authentication, integrity, and resistance to replay/tamper attacks.

Undergraduate Thesis | Harokopio University

Project: Categorization of Cryptographic Vulnerabilities & Web Security Assessment (Grade: 10/10)

- **Tool Development:** Built **Covsaw**, a custom static analysis tool for automated Python security verification.
- **Framework Evaluation:** Conducted comparative benchmarking against industry-standard tools (Semgrep, Bandit) using a corpus of 200 Python files, providing data-driven insights into tool precision and recall.
- **Educational Platform:** Developed **SafeCrypto**, a web platform providing interactive examples and CWE-based documentation for secure coding practices.